

Online Learning

The Multiplicative-Weight Update Algorithm

Joseph Chuang-Chieh Lin

Department of Computer Science & Engineering,
National Taiwan Ocean University

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Credits for the resource

- The slides are based on
 - the lectures of Prof. Luca Trevisan:
<https://lucatrevisan.github.io/40391/index.html>
 - the lectures of Prof. Shipra Agrawal:
<https://ieor8100.github.io/mab/>
 - the lectures of Prof. Francesco Orabona:
<https://parameterfree.com/lecture-notes-on-online-learning/>
 - the monograph: <https://arxiv.org/abs/1912.13213>
 - Elad Hazan's textbook:
Introduction to Online Convex Optimization, 2nd Edition.



Outline

- 1 Expert Setting
- 2 Multiplicative-Weight Update



Listen to the experts?

- Let's say we have n experts.
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- The idea: at each time step, decide the probability distribution (i.e., weights) of the experts to follow their advice.
 - $\mathbf{x}_t = (\mathbf{x}_t(1), \mathbf{x}_t(2), \dots, \mathbf{x}_t(n))$, where $\mathbf{x}_t(i) \in [0, 1]$ and $\sum_i \mathbf{x}_t(i) = 1$.



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- The loss of following expert i at time t : $\ell_t(i)$.
- The expected loss of the algorithm at time t :

$$\langle \mathbf{x}_t, \ell_t \rangle = \sum_{i=1}^n \mathbf{x}_t(i) \ell_t(i).$$



The regret of listening to the experts...

$$\text{regret}_T = \sum_{t=1}^T \langle \mathbf{x}_t, \ell_t \rangle - \min_i \sum_{t=1}^T \ell_t(i).$$

- The set of feasible solutions $K = \Delta \subseteq \mathbb{R}^n$, probability distributions over $\{1, 2, \dots, n\}$.
- $f_t(\mathbf{x}) = \sum_i \mathbf{x}(i) \ell_t(i)$: linear function.
- ★ Assume that $|\ell_t(i)| \leq 1$ for all t and i .
 - Normalized.



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- ★ Assume that $|\ell_t(i)| \leq 1$ for all t and i .
 - Normalized.
- In fact, we claim that (exercise!)

$$\min_i \sum_{t=1}^T \ell_t(i) = \min_{\mathbf{x}} \sum_{t=1}^T \langle \mathbf{x}, \ell_t \rangle.$$



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Multiplicative Weight Update (MWU)

- Maintain a vector of weights $\mathbf{w}_t = (\mathbf{w}_t(1), \dots, \mathbf{w}_t(n))$ where $\mathbf{w}_1 := (1, 1, \dots, 1)$.
- Update the weights at time t by
 - $\mathbf{w}_t(i) := \mathbf{w}_{t-1}(i) \cdot e^{-\beta \ell_{t-1}(i)}$.
 - $\mathbf{x}_t := \frac{\mathbf{w}_t(i)}{\sum_{j=1}^n \mathbf{w}_t(j)}$.

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The weight of expert i at time t : $e^{-\beta \sum_{k=1}^{t-1} \ell_k(i)}$.



Example in portfolio management

Multiplicative Weight Update (MWU)

- ① Maintain a vector of weights $\mathbf{w}_t = (\mathbf{w}_t(1), \dots, \mathbf{w}_t(n))$ representing the ratios of investment for each stock where $\mathbf{w}_1 \leftarrow (1, 1, \dots, 1)$.
- ② Update the weights at time t by
 - ① Derive the PnL $\ell_{t-1}(i)$ of the investment on stock i at time $t - 1$ for each $i = 1, 2, \dots, n$.
 - ② $\mathbf{w}_t(i) \leftarrow \mathbf{w}_{t-1}(i) \cdot e^{-\beta \ell_{t-1}(i)}$. β : a constant parameter
 - ③ $\mathbf{x}_t(i) \leftarrow \frac{\mathbf{w}_t(i)}{\sum_{j=1}^n \mathbf{w}_t(j)}$ for each $i = 1, 2, \dots, n$,
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/* The weight of expert i at time t : $e^{-\beta \sum_{k=1}^{t-1} \ell_k(i)}$.*/



MWU is of no-regret

Theorem 1 (MWU is of no-regret)

Assume that $|\ell_t(i)| \leq 1$ for all t and i . For $\beta \in (0, 1/2)$, the regret of MWU after T steps is bounded as

$$\text{regret}_T \leq \beta \sum_{t=1}^T \sum_{i=1}^n \mathbf{x}_t(i) \ell_t^2(i) + \frac{\ln n}{\beta} \leq \beta T + \frac{\ln n}{\beta}.$$

In particular, if $T > 4 \ln n$, then

$$\text{regret}_T \leq 2\sqrt{T \ln n}$$

by setting $\beta = \sqrt{\frac{\ln n}{T}}$.



Proof of Theorem 1

Let $W_t := \sum_{i=1}^n \mathbf{w}_t(i)$.

- The total weight at time t .

The idea:

- If the algorithm incurs a large loss after T steps, then W_{T+1} is small.
- And, if W_{T+1} is small, then even the best expert performs quite badly.



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Let $L^* := \min_i \sum_{t=1}^T \ell_t(i)$.

- The cumulative loss of the “best” expert.



The proof (contd.)

Lemma 1 (W_{T+1} is SMALL $\Rightarrow L^*$ is LARGE)

$$W_{T+1} \geq e^{-\beta L^*}.$$

Proof.

Let $j = \arg \min_i L^* = \arg \min_i \sum_{t=1}^T \ell_t(i)$.

$$W_{T+1} = \sum_{i=1}^n e^{-\beta \sum_{t=1}^T \ell_t(i)} \geq e^{-\beta \sum_{t=1}^T \ell_t(j)} = e^{-\beta L^*}.$$



The proof (contd.)

Lemma 2 (MWU brings large loss $\Rightarrow W_{T+1}$ is SMALL)

$$W_{T+1} \leq n \prod_{t=1}^n (1 - \beta \langle \mathbf{x}_t, \ell_t \rangle + \beta^2 \langle \mathbf{x}_t, \ell_t^2 \rangle),$$

Proof.

Note: $W_1 = n$.

$$\frac{W_{t+1}}{W_t} = \sum_{i=1}^n \frac{\mathbf{w}_{t+1}(i)}{W_t} = \sum_{i=1}^n \frac{\mathbf{w}_t(i) \cdot e^{-\beta \ell_t(i)}}{W_t}$$

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Hence

$$\ln W_{T+1} \leq \ln n - \left(\sum_{i=1}^T \beta \langle \ell_t, \mathbf{x}_t \rangle \right) + \left(\sum_{i=1}^T \beta^2 \langle \ell_t^2, \mathbf{x}_t \rangle \right)$$

and $\ln W_{T+1} \geq -\beta L^*$ (by Lemma 1).



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Take $\beta = \sqrt{\ln n / T}$, we have $\text{regret}_T \leq 2\sqrt{T \ln n}$.



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Note: $\sum_{i=1}^n \mathbf{x}_t(i) = 1$ and $0 \leq \ell_t^2(i) \leq 1$.



Discussions

